

25.

Ans: A

By Lenz's law, an induced current will flow in the loop such that it opposes the increase in the magnetic flux linkages producing it. When loop starts to enter the region of magnetic field, loop experiences an upwards retarding force larger than its weight and hence decelerate.

Since the loop and the region of magnetic field has the same height d , there will not be a time interval which the entire loop remains in the field.

When the loop is leaving the region of magnetic field, the magnetic flux linkages through the loop decreases, hence the induced current is such that it opposes the decrease in the magnetic flux linkages. → induced current is now in the opposite direction but force is still upwards.

26

Ans: B